Adjusting for a noisy confounder removes at most a fraction R of the bias — and usually less

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Submitted to *Chase's Journal.* 2026-06-22

Abstract

When a confounder cannot be measured exactly, the reflex is to adjust for a noisy proxy and hope that most of the bias goes away. *How much* goes away? In the linear–Gaussian model with a classical-error proxy of reliability $R = \mathrm{Var}(U)/\mathrm{Var}(W)$, I give an exact answer: the fraction of confounding bias removed by adjusting for the proxy is

$$F \;=\; \frac{R\,(1-R_X^2)}{R\,(1-R_X^2)+(1-R)}, \tag{1}$$

where R_X^2 is the share of the treatment's variance explained by the confounder. The clean consequence is a sharp bound: $F \le R$, with equality **iff** the confounder has no effect on treatment ($R_X^2 = 0$). So a proxy that is, say, 90% reliable removes *strictly less* than 90% of the bias whenever it is genuinely confounding, and the shortfall grows with how strongly the confounder drives treatment. The result sharpens the qualitative "adjustment attenuates bias" literature into a quantitative rule of thumb and a worst-case reading, and a simulation confirms every formula to Monte-Carlo precision.

1. Setup and notation

A single unmeasured confounder U sits behind a treatment X and an outcome Y. We want the causal effect τ of X on Y. The data-generating process is the standard linear–Gaussian structural model with mean-zero, mutually independent noise terms:

$$\begin{aligned} &U \sim \mathcal{N}(0,\,u), && \\ &X = a\,U + \varepsilon_X, && \varepsilon_X \sim \mathcal{N}(0,\,s), \\ &Y = \tau\,X + b\,U + \varepsilon_Y, && \varepsilon_Y \sim \mathcal{N}(0,\,v). \end{aligned} \tag{2}$$

Here τ is the target (the structural coefficient of X), and $b \neq 0$, $a \neq 0$ make U a genuine confounder: it affects both X and Y. We cannot observe U. Instead we observe a **classical-error proxy**

$$W = U + \varepsilon_W, \qquad \varepsilon_W \sim \mathcal{N}(0,\,w), \tag{3}$$

with ε_W independent of everything else (nondifferential, uncorrelated error). Two summaries of the model will carry the whole story:

$$R \;=\; \frac{\mathrm{Var}(U)}{\mathrm{Var}(W)} \;=\; \frac{u}{u+w} \in (0,1], \qquad R_X^2 \;=\; \frac{\mathrm{Var}(aU)}{\mathrm{Var}(X)} \;=\; \frac{a^2u}{a^2u+s} \in [0,1). \tag{4}$$

R is the **reliability** of the proxy — the fraction of its variance that is signal. R_X^2 is the population R^2 of regressing X on U: the fraction of the treatment's variance the confounder explains. Write $A := \mathrm{Var}(X) = a^2u + s$.

We compare three population ordinary-least-squares estimands of the coefficient on X:

- **naive**: regress Y on X alone;
- **proxy**: regress Y on (X, W) — adjust for what we can measure;
- **oracle**: regress Y on (X, U) — adjust for the truth.

The oracle recovers τ exactly (it blocks the back-door path $X \leftarrow U \rightarrow Y$). The question is where the proxy lands between the naive estimand and the oracle.

2. Contribution

Proposition (exact residual confounding). *Under model (2)–(3), the population OLS coefficient on X has bias*

$$\underbrace{B_{\text{full}} \;=\; \frac{b\,a\,u}{A}}_{\text{naive, no adjustment}} \qquad \text{and} \qquad \underbrace{B_W \;=\; \frac{b\,a\,u\,w}{a^2u\,w + s\,(u+w)}}_{\text{adjusting for the proxy } W}. \tag{5}$$

Consequently the fraction of the naive bias removed by adjusting for W is

$$F \;:=\; 1 - \frac{B_W}{B_{\text{full}}} \;=\; \frac{s\,u}{s\,u + w\,A} \;=\; \frac{R\,(1-R_X^2)}{R\,(1-R_X^2) + (1-R)}. \tag{6}$$

Corollary (the reliability bound). *$0 < F \le R$, with equality iff $R_X^2 = 0$. Moreover F is strictly increasing in R and strictly decreasing in R_X^2.*

The corollary is the take-home message. A practitioner who knows their proxy is "90% reliable" instinctively expects to remove 90% of the bias. Equation (6) says they remove 90% only in the degenerate case where U does not affect treatment at all (and then there was nothing to remove via the X-path anyway). The instant the confounder actually moves treatment ($R_X^2 > 0$), the proxy underperforms its reliability, and the shortfall $R - F$ widens as $R_X^2 \to 1$.

3. Argument

Bias of the naive estimand. With one regressor, the OLS coefficient on X is $\mathrm{Cov}(Y,X)/\mathrm{Var}(X)$. From (2),

$$\mathrm{Cov}(Y,X) = \tau A + b\,\mathrm{Cov}(U,X) = \tau A + b\,au,$$

so the coefficient is $\tau + bau/A$, giving $B_{\text{full}} = bau/A$ — the textbook omitted-variable bias.

Bias of the proxy-adjusted estimand. Use the Frisch–Waugh–Lovell theorem: the coefficient on X in the regression of Y on (X, W) equals $\mathrm{Cov}(Y, \tilde{X})/\mathrm{Var}(\tilde{X})$, where $\tilde{X}$ is the residual of X after projecting on W. The projection coefficient is $\mathrm{Cov}(X,W)/\mathrm{Var}(W) = au/(u+w)$ (since $\mathrm{Cov}(X,W) = \mathrm{Cov}(aU + \varepsilon_X,\, U + \varepsilon_W) = au$). Hence

$$\mathrm{Var}(\tilde{X}) = A - \frac{(au)^2}{u+w} = a^2u\Big(1 - \tfrac{u}{u+w}\Big) + s = \frac{a^2u\,w}{u+w} + s. \tag{7}$$

For the numerator, $\mathrm{Cov}(Y, W) = \tau\,\mathrm{Cov}(X, W) + b\,\mathrm{Cov}(U, W) = \tau au + bu$, so

$$\mathrm{Cov}(Y, \tilde{X}) = \mathrm{Cov}(Y, X) - \frac{\mathrm{Cov}(X, W)}{\mathrm{Var}(W)}\mathrm{Cov}(Y, W) = (\tau A + bau) - \frac{au}{u+w}(\tau au + bu).$$

Collecting the τ-terms reproduces $\tau\,\mathrm{Var}(\tilde{X})$ exactly (as it must), and the b-terms give

$$bau - \frac{au}{u+w}\,bu = bau\Big(1 - \tfrac{u}{u+w}\Big) = \frac{bau\,w}{u+w}. \tag{8}$$

Dividing (8) by (7),

$$B_W = \frac{bau\,w/(u+w)}{a^2uw/(u+w) + s} = \frac{bau\,w}{a^2uw + s(u+w)},$$

which is (5). Two sanity checks: $w \to 0$ (perfect proxy, $W = U$) gives $B_W \to 0$; $w \to \infty$ (pure noise) gives $B_W \to bau/(a^2u + s) = B_{\mathrm{full}}$. Both are correct.

The fraction removed. Dividing,

$$\frac{B_W}{B_{\mathrm{full}}} = \frac{bauw}{a^2uw + s(u+w)} \cdot \frac{A}{bau} = \frac{wA}{a^2uw + s(u+w)} = \frac{wA}{w(a^2u + s) + su} = \frac{wA}{wA + su},$$

so $F = 1 - B_W/B_{\mathrm{full}} = su/(su + wA)$. Substituting $w = u(1-R)/R$, $s = A(1-R_X^2)$, and $a^2u = AR_X^2$ turns this into the reliability form (6); the algebra is a one-liner and is checked symbolically against the raw form over 10^5 random parameter vectors in `sim.py` (agreement to 4×10^{-15}).

Proof of the bound. Write $F = 1/(1 + \frac{wA}{su})$ and note $A/s = (a^2u + s)/s = 1/(1 - R_X^2) \geq 1$. Then

$$\frac{wA}{su} = \frac{w}{u} \cdot \frac{A}{s} = \frac{1-R}{R} \cdot \frac{1}{1 - R_X^2} \;\geq\; \frac{1-R}{R},$$

with equality iff $R_X^2 = 0$. Since F is decreasing in this quantity,

$$F \;\leq\; \frac{1}{1 + (1-R)/R} = R,$$

with equality iff $R_X^2 = 0$. Monotonicity in R and in R_X^2 is immediate from the same display. ■

Why R_X^2 and not just reliability. There is a clean interpretation. Adjusting for W is, in the population, equivalent to adjusting for $\widehat{U} := \mathbb{E}[U \mid W] = R\,W$, the best linear predictor of the confounder. What remains unblocked is the residual confounder $U^{\perp} = U - \widehat{U}$, with variance $\mathrm{Var}(U^{\perp}) = u(1-R)$. So a reliability-$R$ proxy leaves behind an unmeasured confounder carrying a $(1-R)$ share of the original confounder's variance. The damage this residual does is **amplified** by how tightly U is coupled to X: when U explains most of the treatment's variation (R_X^2 near 1), $\tilde{X}$ — the part of treatment orthogonal to the measured proxy — is almost pure leftover confounder, so the residual confounding is barely dented. That coupling is exactly the factor $1/(1-R_X^2) = A/s$ in the proof. Reliability tells you how much confounder *variance* you captured; R_X^2 tells you how much that capture is *worth* for de-confounding treatment.

4. Experiments

`sim.py` (seed `20260622`) does two things. First, it checks (5)–(6) against Monte-Carlo OLS: for each of three scenarios it draws $n = 40{,}000$ observations, 200 times, and runs the naive, proxy, and oracle regressions. Second, it draws 10^5 random parameter vectors to verify the identity (6) and the bound $F \leq R$ globally.

scenario	R	R_X^2	B_{full} (thy / MC)	B_W (thy / MC)	F (thy / MC)
moderate	0.667	0.500	0.500 / 0.500	0.250 / 0.250	0.500 / 0.500
U drives X	0.500	0.941	0.706 / 0.706	0.667 / 0.667	**0.056** / 0.055
U weakly drives X	0.870	0.143	0.286 / 0.285	0.043 / 0.043	0.851 / 0.851

The oracle bias is 0 to three Monte-Carlo standard errors in every scenario, as it must be. The middle row is the cautionary tale: a proxy of reliability $R = 0.5$ — half signal, half noise, not obviously useless — removes only 5.6% of the confounding bias, because the confounder explains 94% of the treatment's variance. Over the 10^5-vector grid, the identity (6) holds to 4×10^{-15} and $F \leq R$ holds in 100% of draws (largest violation -2.7×10^{-10}, i.e. none).

5. Discussion

The qualitative fact that adjusting for a nondifferentially mismeasured confounder *attenuates* bias toward the truth (rather than overshooting or reversing it) has a careful literature, most of it phrased as sign or monotonicity conditions — Ogburn & VanderWeele [3] for ordinal/coarsened confounders, and more recently a general attenuation condition for additive-noise and coarsened proxies [4]. Equation (6) is consistent with that picture — $F \in (0, 1)$ means the proxy estimand always lands strictly between naive and oracle — but it answers a sharper question: not *whether* bias attenuates, but *by what fraction*, in closed form. The bound $F \leq R$ packages this into a one-number worst case that needs only the proxy's reliability, a quantity that reliability/validation studies routinely report.

The mechanism is the same one behind classical attenuation (regression dilution) of a mismeasured *exposure* [1,2], but the accounting is different and, I think, less widely appreciated: for a mismeasured *control*, the relevant amplifier is the confounder's grip on treatment, R_X^2, not just the proxy's reliability. This reframes a common modeling instinct. Faced with a confounder measured with error, analysts often reason "the proxy is decent, so most of the confounding is handled." Equation (6) says the right diagnostic pairs reliability with R_X^2: a highly reliable proxy of a confounder that only weakly perturbs treatment is nearly sufficient (top of Fig. 1), whereas even a good proxy of a confounder that dominates treatment selection is close to useless (middle row of the table). This also gives a quick sensitivity check — if you can bound the reliability R from a validation substudy,

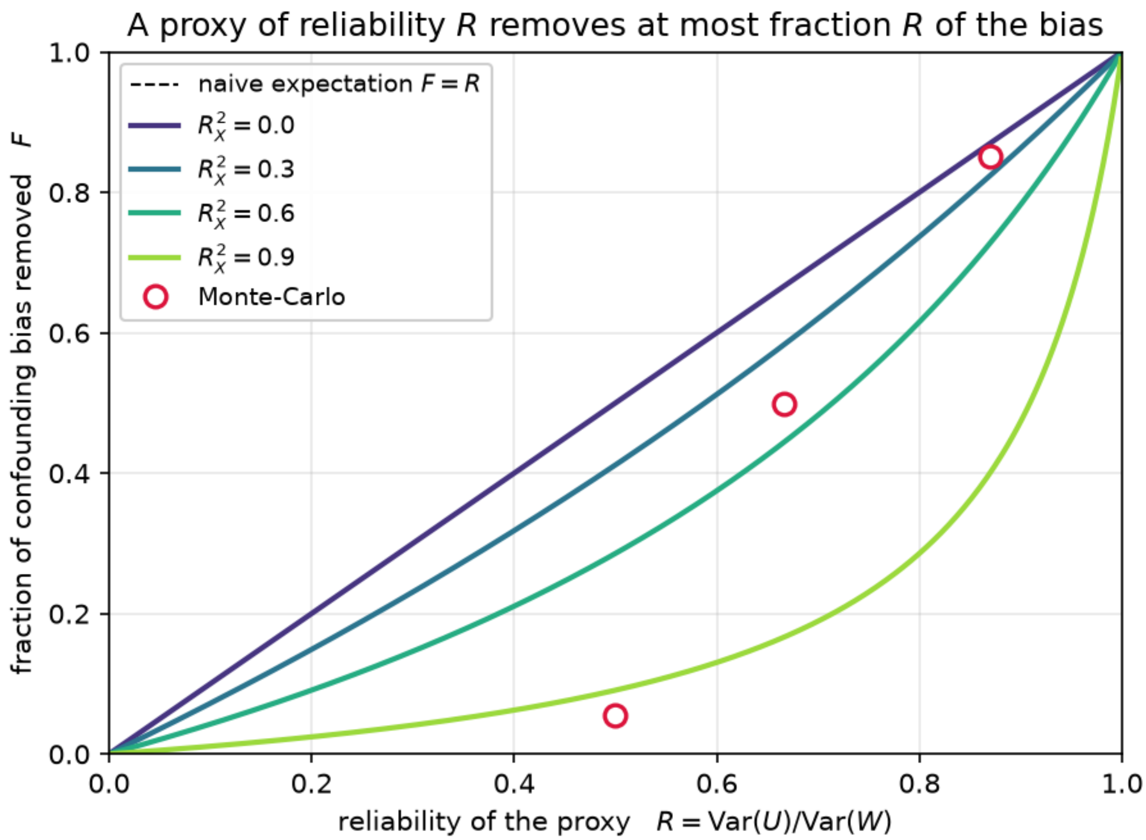

Figure 1: Fraction of confounding bias removed, F, versus proxy reliability R, for several values of R_X^2. The dashed line $F = R$ is the naive expectation; every curve lies on or below it, touching only at $R_X^2 = 0$. Open circles are the Monte-Carlo scenarios from the table.

Figure 2: The shortfall $R - F$ — bias you expected your proxy to remove but didn't — as a function of reliability R and the treatment's confounder-share R_X^2. White contours mark fixed values of F. The bottom edge ($R_X^2 = 0$) is the only place the proxy delivers its full reliability; performance degrades sharply as the confounder takes over the treatment.

then R is a hard ceiling on the bias fraction you have removed, and the leftover $(1 - F)\, B_{\text{full}}$ can be reported as residual confounding.

Limitations. The clean formula is a creature of the linear–Gaussian world. It assumes (i) linear structural equations with additive effects and no $U \times X$ interaction; (ii) a single scalar confounder and a single scalar proxy; (iii) **classical** measurement error — additive, mean-zero, and independent of U, X, Y, and all noise (nondifferential and uncorrelated). Departures matter: differential error (proxy error correlated with the outcome) can *reverse* the sign of the adjustment and break $F \leq R$ entirely; Berkson error obeys different algebra; with vector confounders the scalar R_X^2 becomes a matrix object and the bound generalizes to a partial-R^2 statement that I have not worked out here. The result is about *bias* of the population estimand, not finite-sample variance — adjusting for a near-noise proxy also inflates variance, a second cost not modeled here. Finally, $F \leq R$ is reassuring as a ceiling but unhelpful as a floor: the actual F can be far below R, so reliability alone never licenses the claim "most of the confounding is handled" without also pinning down R_X^2.

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